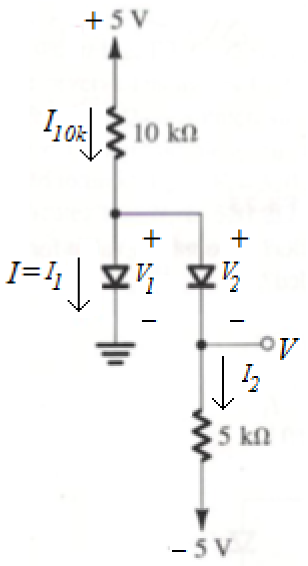
3.9 Assuming that the diodes in the circuits of Fig. P3.9 are ideal, find the values of the labeled voltages and currents.



Assume that both D1 and D2 are cut off:

In that case, $I_{10k\Omega} = 0 \Rightarrow V_1 = 5$ V **!** Also, $V_2 = 5 - (-5) = 10$ V **!**

but both V_1 and V_2 must be $< 0 \Rightarrow$ our assumption is incorrect.

Assume that both D1 and D2 are turned on:

In that case, $I_{10k\Omega} = \frac{(5-0)}{(10\text{ k})} = 0.5$ mA. $I_2 = \frac{(0-(-5))}{(5\text{ k})} = 1$ mA $\Rightarrow I_1 = I_{10k\Omega} - I_2 = -0.5$ mA **!**

$\because I_1 < 0$ which contradicts the condition of a turned-on diode $\Rightarrow$ our assumption is incorrect.

Assume that D1 is on and D2 is off:

In that case, $I_{10k\Omega} = I_1 = \frac{(5-0)}{(10\text{ k})} = 0.5$ mA. $V_2 = 0 - (-5) = 5$ V **!**

$\because V_2 > 0 \Rightarrow$ our assumption is incorrect.

Assume that D1 is off and D2 is on:

In that case, $I_{10k\Omega} = I_2 = \frac{(5-(-5))}{(15\text{ k})} = (2/3)$ mA. $V_1 = [5 - (10\text{ k}) \times (2/3\text{ m})] - 0 = -1.667$ V ✓

$\because V_1 < 0$ and $I_2 > 0$ that confirms our assumption. ☺

$\Rightarrow I = I_1 = 0$; and $V = V_1 = (5\text{ k}) \times (2/3\text{ m}) + (-5) = -1.667$ V

Let us resolve the above problem using a constant voltage drop model with V_D = 0.7 V;

Assume that both D1 and D2 are cut off:

In that case, $I_{10k\Omega} = 0 \Rightarrow V_1 = 5$ V **!** Also, $V_2 = 5 - (-5) = 10$ V **!**

$\Rightarrow$ our assumption is incorrect.

Assume that both D1 and D2 are turned on:

In that case, $I_{10k\Omega} = \frac{(5 - 0.7)}{10\text{ k}} = 0.43$ mA. $I_2 = \frac{(0.7 - 0.7 - (-5))}{5\text{ k}} = 1.0$ mA $\Rightarrow I_1 = I_{10k\Omega} - I_2 = -0.57$ mA **!**

$\because I_1 < 0 \Rightarrow$ our assumption is incorrect.

Assume that D1 is on and D2 is off:

In that case, $I_{10k\Omega} = I_1 = \frac{5 - 0.7}{10\text{ k}} = 0.43$ mA. $V_2 = 0.7 - (-5) = 5.7$ V **!**

$\because V_2 > 0.7 \Rightarrow$ our assumption is incorrect.

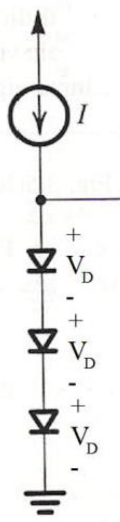
Assume that D1 is off and D2 is on:

In that case, $I_{10k\Omega} = I_2 = \frac{(5 - 0.7 - (-5))}{15\text{ k}} = 0.62$ mA. $V_1 = [5 - (10\text{ k}) \times (0.62\text{ m})] - 0 = -1.2$ V ✓

$\because V_1 < 0.7$ and $I_2 > 0$ that confirms our assumption. ☺

$\Rightarrow I = I_1 = 0$; and $V = (5\text{ k}) \times (0.62\text{ m}) + (-5) = -1.9$ V

3.23 The circuit in Fig. P3.23 utilizes three identical diodes having $n = 1$ and $I_S = 10^{-14}$ A. Find the value of the current I required to obtain an output voltage $V_O = 2$ V. If a current of 1 mA is drawn away from the output terminal by a load, what is the change in output voltage?



∵ the three diodes are identical and conduct the same current (I),

The voltage drop across each diode will be the same = V_D.

Now we have $3V_D = 2 \Rightarrow V_D \approx 0.667$ V or 667 mV.

$\Rightarrow I = 1\times10^{-14} \times e^{\frac{667}{1\times25.8}} \approx 1.69$ mA

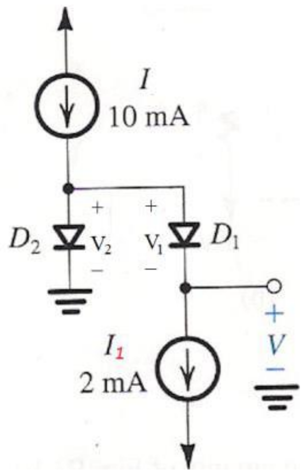
If a current of 1 mA is drawn away from the output terminal that leaves the diodes with 0.69 mA.

The change in a single diode voltage will be $\Delta V_D = 2.3 \times n \times V_T \times \log\frac{0.69}{1.69} = -23.34\ mV$

$\Rightarrow \Delta V = 3\ \Delta V_D = -70$ mV

i.e., $V_{\text{loaded}} = V + \Delta V = 1930$ mV = 1.93 V.

3.25 In the circuit shown in Fig. P3.25, both diodes have $n = 1$, but D_1 has 10 times the junction area of D_2. What value of V results? To obtain a value for V of 50 mV, what current I_1 is needed?



In general,

$$\frac{I_2}{I_1} = \frac{I_{s2}}{I_{s1}} e^{\frac{(V_2-V_1)}{nV_T}}$$

$$V_2 - V_1 = nV_T \ln\left(\frac{I_2}{I_1} \times \frac{I_{s1}}{I_{s2}}\right)$$

$$V_2 - V_1 = 2.3nV_T \log\left(\frac{I_2}{I_1} \times \frac{I_{s1}}{I_{s2}}\right)$$

For this particular problem, $A_{D1} = 10\ A_{D2} \Rightarrow I_{S1} = 10\ I_{S2} \Rightarrow \frac{I_{S1}}{I_{S2}} = 10$

Also by inspection $I_2 = I - I_1 = 8$ mA.

Writing a node equation starting from the cathode of D_2 and going clockwise,

$V_2 - V_1 - V = 0 \Rightarrow V = V_2 - V_1 = 60 \times \log(\frac{8}{2}\times10)$ mV = 96 mV.

$50 = 60 \times \log(\frac{10-I_1}{I_1}\times10) \Rightarrow \log(\frac{10-I_1}{I_1}\times10) = 0.8334 \Rightarrow 10^{0.8334} = \frac{100}{I_1} - 10 \Rightarrow \frac{100}{I_1} = 17.814$

$$\Rightarrow I_1 = 5.61\ mA$$

3.27 Several diodes having a range of sizes, but all with $n = 1$, are measured at various temperatures and junction currents as noted below. For each, estimate the diode voltage at 1 mA and 25°C.

(a) 620 mV at 10 μA and 0°C
(b) 790 mV at 1 A and 50°C
(c) 590 mV at 100 μA and 100°C
(d) 850 mV at 10 mA and –50°C
(e) 700 mV at 100 mA and 75°C

a)

$$V|_{T=25^\circ C, I=10\mu A} = V|_{T=0^\circ C, I=10\mu A} + (25-0)\times(-2\text{ mV})$$
$$V|_{T=25^\circ C, I=10\mu A} = 620\text{ mV} + (-50\text{ mV})$$
$$V|_{T=25^\circ C, I=10\mu A} = 570\text{ mV}$$

$$V|_{T=25^\circ C, I=1mA} = V|_{T=25^\circ C, I=10\mu A} + 2.3nV_T\log\left(\frac{1\times10^{-3}}{10\times10^{-6}}\right)$$
$$V|_{T=25^\circ C, I=1mA} = (570+120)\text{ mV} = 690\text{ mV} \Leftarrow$$

b)

$$V|_{T=25^\circ C, I=1A} = V|_{T=50^\circ C, I=1A} + (25-50)\times(-2\text{ mV})$$
$$V|_{T=25^\circ C, I=1A} = 790\text{ mV} + 50\text{ mV}$$
$$V|_{T=25^\circ C, I=1A} = 830\text{ mV}$$ because the current is very large!

$$V|_{T=25^\circ C, I=1mA} = V|_{T=25^\circ C, I=1A} + 2.3nV_T\log\left(\frac{1\times10^{-3}}{1}\right)$$
$$V|_{T=25^\circ C, I=1mA} = (830-180)\text{ mV} = 650\text{ mV} \Leftarrow$$

3.34 A "1-mA diode" (i.e., one that has $v_D = 0.7$ V at $i_D = 1$ mA) is connected in series with a 200-Ω resistor to a 1.0-V supply.

(a) Provide a rough estimate of the diode current you would expect.
(b) If the diode is characterized by $n = 2$, estimate the diode current more closely using iterative analysis.

a)

$$I_2 = \frac{V_S - V_1}{R} = \frac{1-0.7}{200} = 1.5 \text{ mA}$$

b)

$$V_2 = V_1 + 2.3nV_T \log(\frac{I_2}{I_1}) = 0.7 + 0.119 \times \log(\frac{1.5}{1}) = 0.72095$$

$$I_3 = \frac{V_S - V_2}{R} = \frac{1-0.72095}{200} = 1.39525 \text{ mA}$$

$$V_3 = V_2 + 2.3nV_T \log(\frac{I_3}{I_2}) = 0.72095 + 0.119 \times \log(\frac{1.39525}{1.5}) = 0.71721$$

$$I_4 = \frac{V_S - V_3}{R} = \frac{1-0.71721}{200} = 1.41395 \text{ mA}$$

$$V_4 = V_3 + 2.3nV_T \log(\frac{I_4}{I_3}) = 0.71721 + 0.119 \times \log(\frac{1.41395}{1.39525}) = 0.71789$$

$$I_5 = \frac{V_S - V_4}{R} = \frac{1-0.71789}{200} = 1.41055 \text{ mA}$$

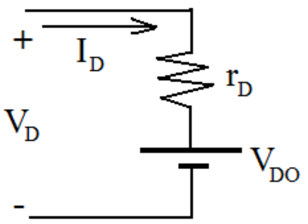
3.37 Find the parameters of a piecewise-linear model of a diode for which $v_D = 0.7$ V at $i_D = 1$ mA and $n = 2$. The model is to fit exactly at 1 mA and 10 mA. Calculate the error in millivolts in predicting v_D using the piecewise-linear model at $i_D = 0.5$, 5, and 14 mA.

$$\left| \Delta V_D \big/ decade(\Delta I_D) \right| = 2.3 n V_T$$

$$\left| \Delta V_D \big/ decade(\Delta I_D) \right| = 0.12 \text{ V/decade @ RT for n = 2}$$

I_D (mA)	V_D (V)
1	0.7
10	0.7+0.12=0.82

$$r_D = \frac{\Delta V_D}{\Delta I_D} = \frac{0.12}{(10-1)\times 10^{-3}} = 13.34\ \Omega$$



$$V_D = I_D \times r_D + V_{DO} \Rightarrow V_{DO} = V_D - I_D \times r_D$$

$$V_{DO} = 0.7 - 1\times 10^{-3} \times 13.34 = 0.687 \text{ V}$$

To calculate V_D exactly, we need to find I_S :

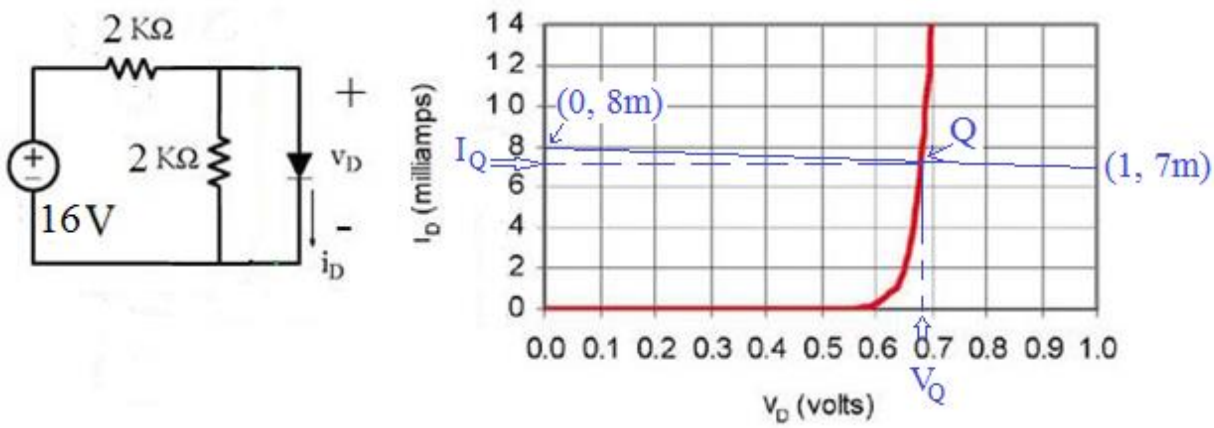
$$I_S = I_D \times e^{-(\frac{V_D}{nV_T})} = 1\times 10^{-3} \times e^{-(\frac{0.7}{2\times 0.0258})} = 1.2835\times 10^{-9} \text{ A}$$

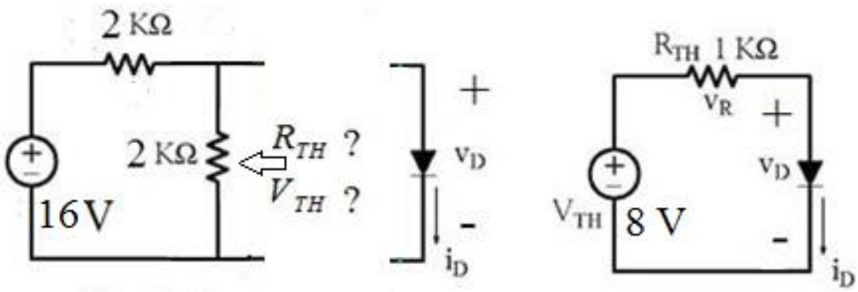
I_D (mA)	$V_D = nV_T \times \ln(\frac{I}{I_S}+1)$	V_D using Piecewise-linear model	Difference (mV)
0.5	0.6642	0.6937	29.5
5	0.7831	0.7537	-29.4
14	0.8362	0.8738	37.6

Graphical Analysis (Load Line Method):

- Example 1:

The diode used in the circuit below has the i-v characteristic shown to the right. Find the the dc operating (bias) point Q.





Load Line Equation

$$V_{TH} = I_D \times R_{TH} + V_D$$

$$8 = I_D \times 1000 + V_D$$

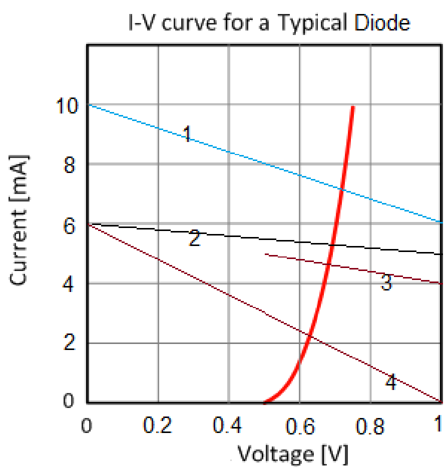
$$@V_D = 0\,\text{V} \rightarrow I_D = 8\text{ mA}$$

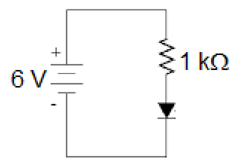
$$@V_D = 1\,\text{V} \rightarrow I_D = 7\text{ mA}$$

- Example 2

In the given i-v characteristic below, which load line represent the given circuit?

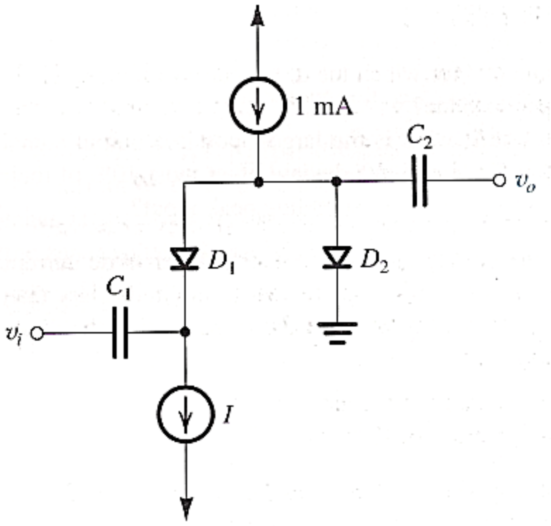
Answer: Number 2.





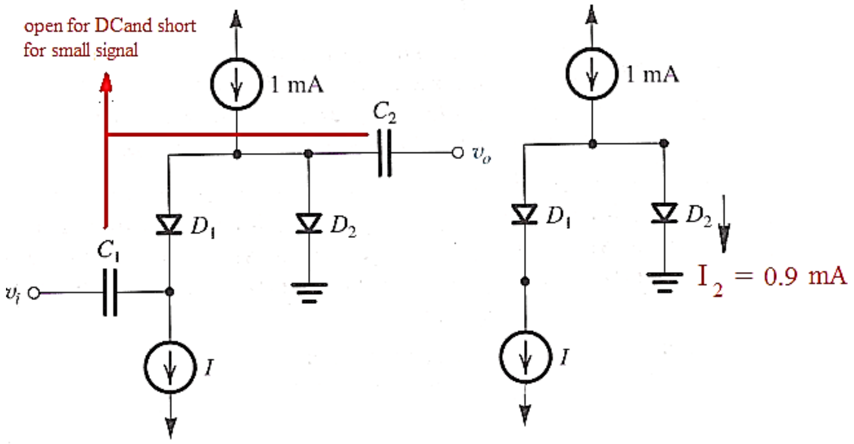
3.56 In the capacitor-coupled attenuator circuit shown in Fig. P3.56, I is a dc current that varies from 0 mA to 1 mA, D_1 and D_2 are diodes with $n = 1$, and C_1 and C_2 are large coupling capacitors. For very small input signals, find the values of the ratio v_o/v_i for I equal to:

(a) 0 μA
(b) 1 μA
(c) 10 μA
(d) 100 μA
(e) 500 μA
(f) 600 μA
(g) 900 μA
(h) 990 μA
(i) 1 mA



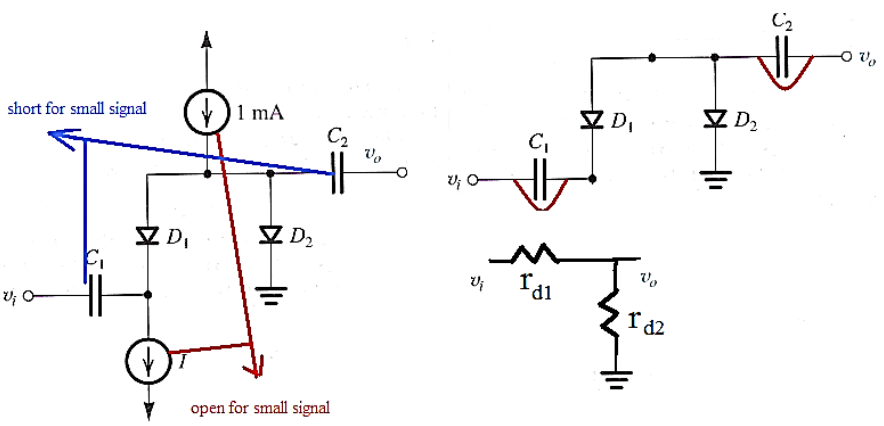
d) I=100 µA

DC Equivalent circuit:



Small Signal Equivalent circuit:

$$r_{d1} = \frac{nV_T}{I_1} = \frac{1\times.0259}{100\times10^{-6}} = 259\ \Omega \qquad r_{d2} = \frac{nV_T}{I_2} = \frac{1\times.0259}{0.9\times10^{-3}} = 28.78\ \Omega$$



$$v_o/v_i = \frac{r_{d2}}{r_{d2}+r_{d1}} = \frac{28.78}{28.78+259} = 0.1\ V/V$$

3.107)

A young designer, aiming to develop intuition, concerning conducting paths within an integrated circuit, examines the end-to-end resistance of a connecting bar 10 μm long, 3 μm wide, 1 μm thick, made of various materials. The designer considers:

a) intrinsic silicon
b) *n*-doped silicon with $N_D = 1 \times 10^{16}/\text{cm}^3$
c) *n*-doped silicon with $N_D = 1 \times 10^{18}/\text{cm}^3$
d) *p*-doped silicon with $N_D = 1 \times 10^{10}/\text{cm}^3$
e) aluminum with resistivity of 2.8 μΩ·cm

Find the resistance in each case. For intrinsic silicon, use $\mu_n = 1350$ cm²/V·s, $\mu_p = 480$ cm²/V·s, and $n_i = 1.5 \times 10^{10}/\text{cm}^3$. For doped silicon, assume $\mu_n \cong 2.5\mu_p = 1200$ cm²/V·s. (Recall that $R = \rho L/A$.)

In general,

$$n = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \quad ;\text{and} \quad p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2}$$

$$a)\ n = p = n_i \quad \Rightarrow R = \frac{\rho L}{A} = \frac{L}{\sigma A} = \frac{L}{[q(n\mu_n + p\mu_p)]A} = \frac{L}{[qn_i(\mu_n + \mu_p)]A} \Rightarrow$$

$$R = \frac{10 \times 10^{-4}}{1.602 \times 10^{-19} \times 1.5 \times 10^{10}(1350+480) \times 3 \times 10^{-4} \times 1 \times 10^{-4}} = 7.58 \times 10^9\ \Omega = 7.59\ \text{G}\Omega$$

$$b)\ n = N_D; p = \frac{n_i^2}{N_D} \Rightarrow R = \frac{\rho L}{A} = \frac{L}{\sigma A} = \frac{L}{q\left(N_D\mu_n + \frac{n_i^2}{N_D}\mu_p\right)A} \approx \frac{L}{q(N_D\mu_n)A}$$

$$(note: N_D\mu_n \gg \frac{n_i^2}{N_D}\mu_p\ ) \quad \Rightarrow$$

$$R = \frac{10 \times 10^{-4}}{1.602 \times 10^{-19} \times 1 \times 10^{16} \times 1200 \times 3 \times 10^{-4} \times 1 \times 10^{-4}} = 17361\ \Omega = 17.361\ k\Omega$$

$d)\ Since\ N_A$ is comparable to n_i,

$$p = \frac{N_A - N_D}{2} + \sqrt{\left(\frac{N_A - N_D}{2}\right)^2 + n_i^2} = 2.08 \times 10^{10}\ cm^{-3}; \qquad n = \frac{n_i^2}{p} = 1.08 \times 10^{10}\ cm^{-3}$$

$$R = \frac{\rho L}{A} = \frac{L}{\sigma A} = \frac{L}{q(n\mu_n + p\mu_p)A}$$

$$R = \frac{10 \times 10^{-4}}{1.602 \times 10^{-19}(2.08 \times 10^{10} \times 480 + 1.08 \times 10^{10} \times 1200) \times 3 \times 10^{-4} \times 1 \times 10^{-4}} = 9.07\ G\Omega$$

$$e)\ \ R = \frac{\rho L}{A} = \frac{2.8 \times 10^{-6} \times 10 \times 10^{-4}}{3 \times 10^{-4} \times 1 \times 10^{-4}} = \ 0.0934\ \ \Omega = 93.4\ m\Omega$$

3.108)

Holes are being steadily injected into a region of n-type silicon. In the Steady state, the excess-hole concentration profile shown in Fig P3.108 Is established in the n-type silicon region. Here "excess" means over and above the concentration p_{no}.

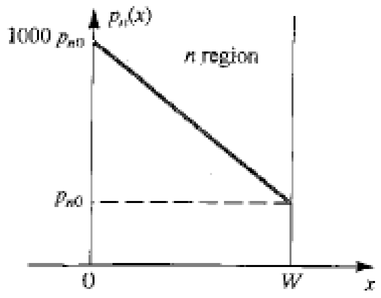


FIGURE P3.108

If $N_D = 10^{16}\ /cm^3$, $n_i = 1.5 \times 10^{10}\ /cm^3$ and $W = 5\mu m$,
find the density of the current that will flow in the x direction.

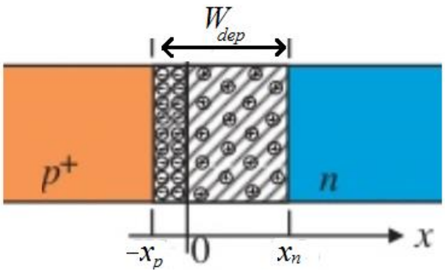
$$J_p = -qD_p\frac{dp}{dx}$$

$$p_{no} = \frac{{n_i}^2}{n_{no}} = \frac{{n_i}^2}{N_D} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4\ cm^{-3}$$

$$\frac{dp}{dx} = \frac{p_{no} - 1000p_{no}}{W - 0} = \frac{-999p_{no}}{W} = \frac{-999 \times 2.25 \times 10^4}{5 \times 10^{-4}} = -4.4955 \times 10^{10}\ cm^{-4}$$

$$J_p = -qD_p\frac{dp}{dx} = -1.602 \times 10^{-19} \times 12 \times (-4.4955 \times 10^{10}) = 8.64 \times 10^{-8}\ A/cm^2$$

3.115)



If for a particular junction, the acceptor concentration is $1 \times 10^{16}/\text{cm}^3$ and the donor concentration is $1 \times 10^{15}/\text{cm}^3$, find the junction built-in voltage (barrier voltage). Assume $n_i = 1 \times 10^{10}/\text{cm}^3$. Also, find W_{dep} and its extent in each of the p and n regions when the junction is reverse biased with $V_R = 5$ V. At this value of the reverse bias, calculate the *magnitude* of the charge stored on either side of the junction. Assume the junction area is 400 μm². Also, calculate C_j.

$$V_O = V_T \times \ln\frac{N_A \times N_D}{{n_i}^2}$$

$$V_O = 0.0258 \times \ln\frac{10^{16} \times 10^{15}}{(10^{10})^2}$$

$$V_O = 0.653\ V$$

$$W_{dep} = \sqrt{\frac{2\varepsilon_s}{q}(\frac{1}{N_A} + \frac{1}{N_D})(V_O + V_R)}$$

$$W_{dep} = \sqrt{\frac{2\times 11.7\times 8.854\times 10^{-14}}{1.602\times 10^{-19}}\left(\frac{1}{10^{16}} + \frac{1}{10^{15}}\right)(0.653+5)} = 2.836 \times 10^{-4}\ \text{cm} = 2.836\ \mu\text{m}$$

$$W_{dep} = x_p + x_n \rightarrow x_p + x_n = 2.836 \quad ① \quad ; \quad \frac{x_p}{x_n} = \frac{N_D}{N_A} \rightarrow \frac{x_p}{x_n} = \frac{10^{15}}{10^{16}} \rightarrow \frac{x_n}{x_p} = 10 \quad ②$$

Solving ① and ② simultaneously yields:

$x_p = \frac{2.836}{11} = 0.258\ \mu\text{m}$ and $x_n = 2.578\ \mu\text{m}$

$$q_j = qN_Dx_nA = qN_Ax_pA = 1.602 \times 10^{-19} \times 10^{16} \times 0.2578 \times 10^{-4} \times 400 \times 10^{-8} = 1.652 \times 10^{-13}\ C$$

$$C_j = \frac{\varepsilon_s \times A}{W_{dep}} = \frac{11.7 \times 8.854 \times 10^{-14} \times 400 \times 10^{-8}}{2.836 \times 10^{-4}} = 1.461 \times 10^{-14}\,F = 14.61\,fF$$

3.121)

A p^+-n diode is one in which the doping concentration in the p region is much greater than that in the n region. In such a diode, the forward current is mostly due to the hole injection across the junction. Show that:

$$I \approx I_p = Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}\right)\left(e^{\frac{V}{V_T}}-1\right)$$

For the specific case in which $N_D = 5\times10^{16}/cm^3$, $D_p = 10\ cm^2/s$, $\tau_p = 0.1\ \mu s$, and $A = 10^4\ \mu m^2$, find I_S and the voltage V obtained when $I = 0.2\ mA$. Assume operation at 300 K where $n_i = 1.5\times10^{10}/cm^3$. Also, calculate the excess minority-carrier charge and the value of the diffusion capacitance at $I = 0.2\ mA$.

$$I = I_S\left(e^{\frac{V}{V_T}}-1\right) = I_p + I_n$$

Where $I_S = Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}+\frac{D_n}{L_nN_A}\right) \Rightarrow I = Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}+\frac{D_n}{L_nN_A}\right)\left(e^{\frac{V}{V_T}}-1\right)$

$$I_p = Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}\right)\left(e^{\frac{V}{V_T}}-1\right)\ and\ \ I_n = Aqn_i^{\,2}\left(\frac{D_n}{L_nN_A}\right)\left(e^{\frac{V}{V_T}}-1\right)$$

$\because$ the junction is p+-n $\Rightarrow N_A \gg N_D \Rightarrow I_p \gg I_n \Rightarrow I \approx I_p$

$$I_p = Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}\right)\left(e^{\frac{V}{V_T}}-1\right) \approx I = 0.2\ mA$$

$$L_p = \sqrt{D_p\tau_p} = \sqrt{10\times0.1\times10^{-6}} = 1\times10^{-3}\ cm$$

$$I_S \approx Aqn_i^{\,2}\left(\frac{D_p}{L_pN_D}\right) = 10^4\times10^{-8}\times1.602^{-19}\times(1.5\times10^{10})^2\frac{10}{1\times10^{-3}\times5\times10^{16}} = 0.72\times10^{-15}\ A$$

$$V \approx V_T\ln(\frac{I_p}{I_S}+1) = 0.0258\times26.35 = 0.6798\ V$$

$$Q_p = \tau_pI_p = 0.1\times10^{-6}\times0.2\times10^{-3} = 2\times10^{-11}\ C = 20\ pC$$

$$C_d = \frac{Q}{V_T} = \frac{Q_p+Q_n}{V_T} \approx \frac{Q_p}{V_T} = \frac{\tau_pI_p}{V_T} = 7.752\times10^{-10} = 0.7752\ nF$$